DocIntel: submission deliverables

AI Engineer assessment, Care Guide. Project: AI-Powered Document Intelligence System. Submitted 7 November 2026.

Web app	https://docintel-tan-two.vercel.app (the login is in the submission email)
API docs	https://docintel-api-y25b.onrender.com/api/docs
Repository	https://github.com/Sowan3k/AI-Powered-Document-Intelligence-System
Walkthrough video	Link added before submission
This report as a PDF	DocIntel-Deliverables.pdf

DocIntel reads invoices and receipts from PDFs, scans, phone photos, TIFF images, Excel workbooks and CSV files. It extracts the invoice fields with Gemini, checks every number in Python, sends doubtful records to a person, cross-checks one document against another, and makes everything searchable by keyword and by meaning.

The brief lists nine deliverables. This page gives each one in short, with the key results, and links to the full report behind it. The backend runs on Render's free plan and sleeps when idle, so the first request can take up to a minute.

1. Git repository with the source code

https://github.com/Sowan3k/AI-Powered-Document-Intelligence-System

Path	Holds
`backend/app/ingest`	File type detection by bytes, PDF, OCR, image and spreadsheet readers
`backend/app/extract`	Gemini client, prompts, schemas, page windows, parsing, evidence checks
`backend/app/validate`	Rules R01 to R11, duplicates, reconciliation
`backend/app/search`	Full-text and vector search
`backend/app/worker`	Job queue in PostgreSQL, pipeline, worker process
`backend/app/api`	REST API (FastAPI)
`frontend/`	Next.js web app
`eval/`	Labelled dataset generator, evaluation runner, load test
`docs/`	Architecture, schema, security, evaluation, test report, AI usage

Every design choice, 85 in all, is in DECISIONS.md with the alternatives, the reason and the trade-off.

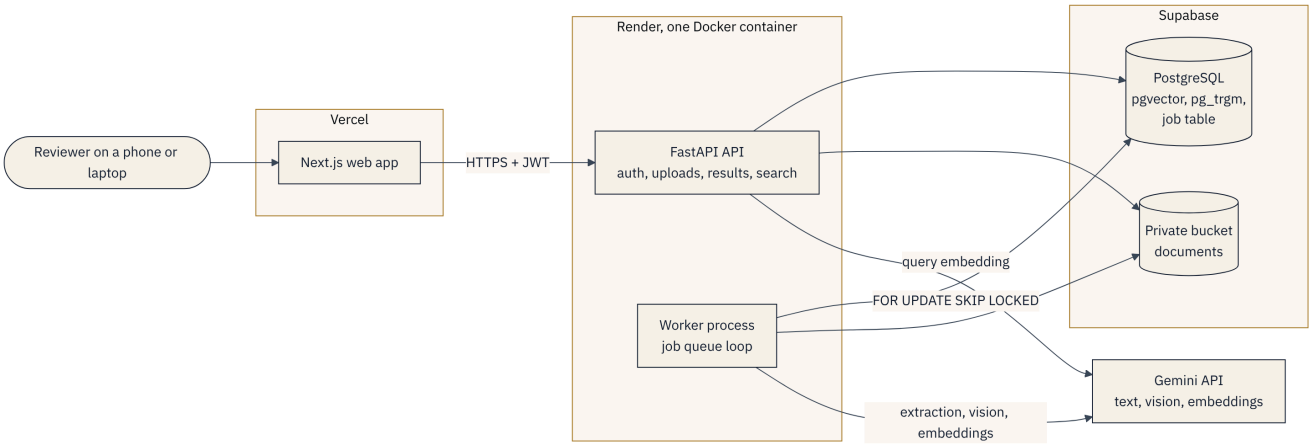
2. README and setup instructions

README.md describes the system, and its Installation and setup section gives two ways to run it. The quick one needs Git and Docker:

```
git clone https://github.com/Sowan3k/AI-Powered-Document-Intelligence-System.git
cd AI-Powered-Document-Intelligence-System
# create .env with GEMINI_API_KEY and the three model names (see the README)
docker compose up --build
```

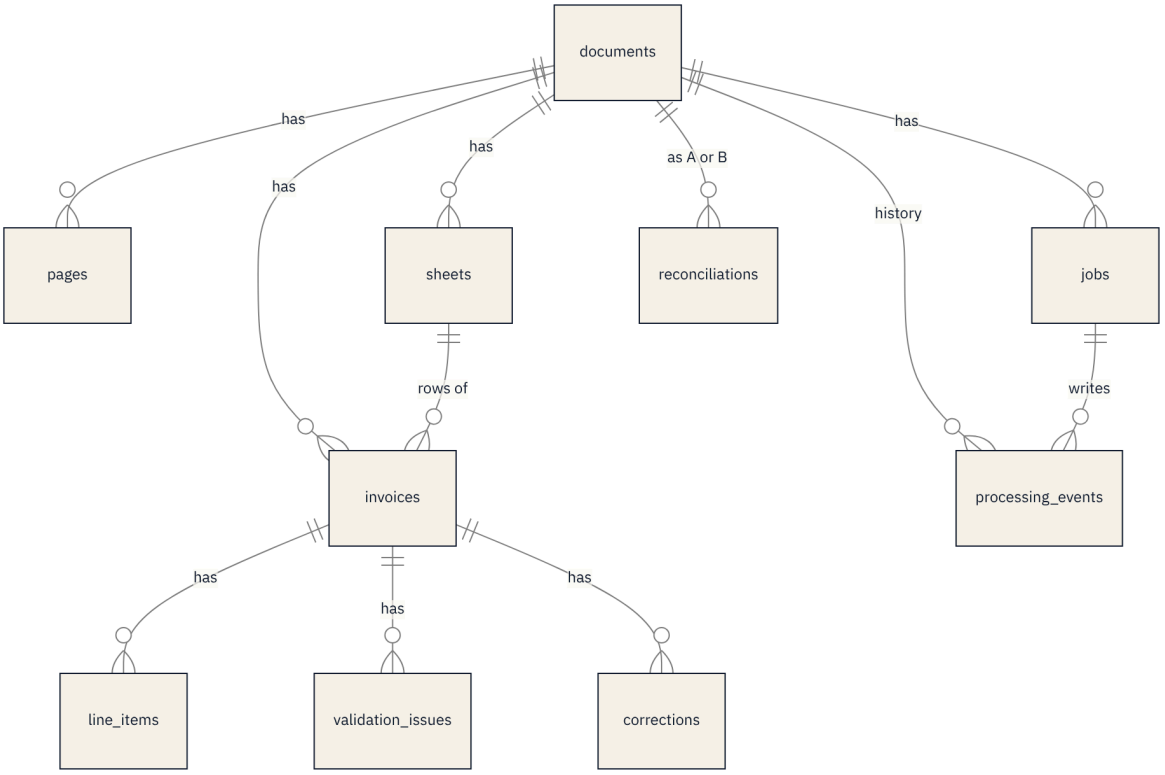
This starts Postgres with pgvector, the API, the worker and the web app at http://localhost:3000. The other way runs each part by hand. Every setting is listed in the README and in `.env.example`.

3. Architecture diagram and database schema



An upload is checked by its bytes, stored in a private bucket under a random key, and queued. The API answers at once. A worker claims the job from a PostgreSQL table and reads the file. It sends the text to Gemini in windows of eight pages, so an invoice that crosses a page edge stays whole. Python then parses the values, runs the rules and indexes the invoice for search. Every step writes a row to the processing history, which the web app shows live.

The database has ten tables:



Money is `NUMERIC(14,2)` in the database and `Decimal` in Python, never a float. Full detail: ARCHITECTURE.md (system and pipeline diagrams) and SCHEMA.md (every column, index and constraint).

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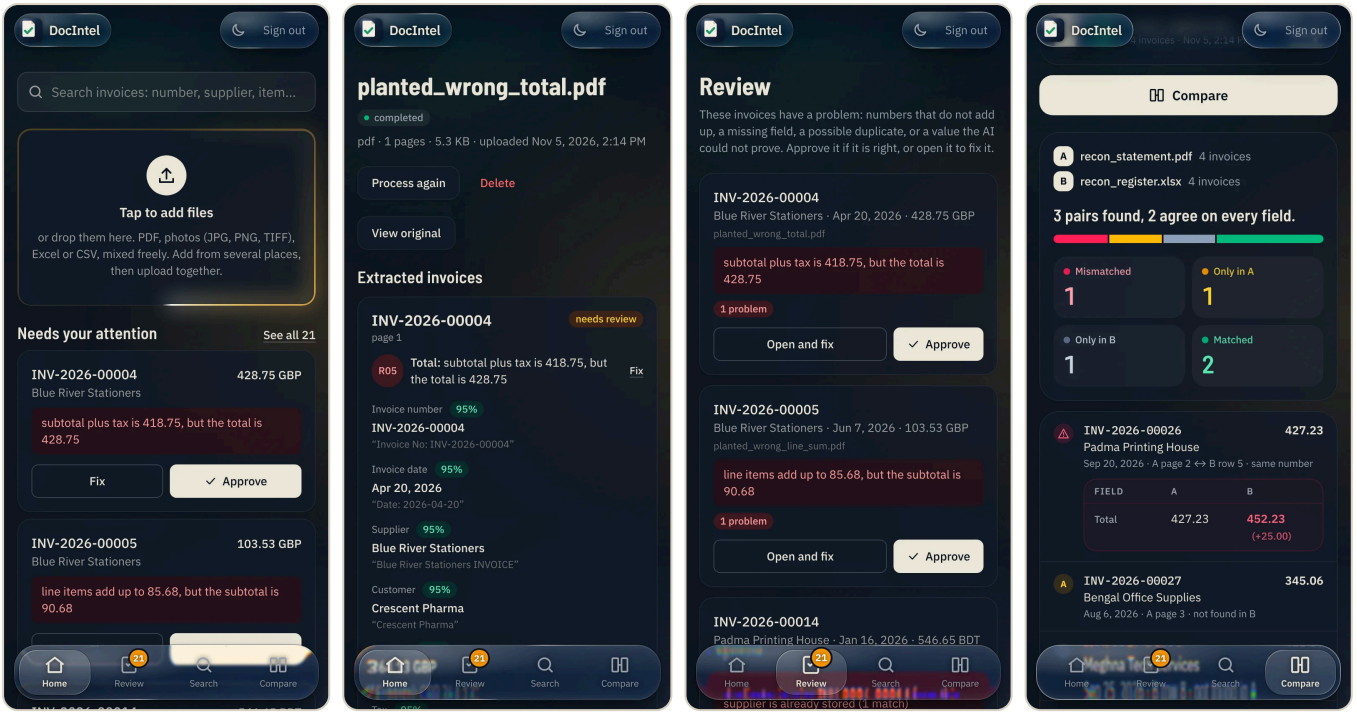
4. Working backend API and frontend

Both are live and deploy from `main`: the web app on Vercel, the API and worker on Render, the database and files on Supabase.

Brief asks for	Endpoint
Document uploads	`POST /api/documents` (up to 20 files at once)
Processing status	`GET /api/documents/{id}`, `GET /api/activity`, `GET /api/jobs/{id}`
Extracted results	`GET /api/documents/{id}/results`, `GET /api/export/invoices.csv`
Document search	`GET /api/search` (keyword, meaning, or both, with filters)
Manual corrections	`PATCH /api/invoices/{id}`, `GET /api/review-queue`
Failed-job reprocessing	`POST /api/jobs/{id}/retry`, `POST /api/documents/{id}/reprocess`
Retake a blurred photo	`POST /api/documents/{id}/replace` (the same document, read again from the new file)
Cross-verification	`POST /api/reconcile`, `GET /api/reconcile/suggestions`
The original file	`GET /api/documents/{id}/file`, `GET /api/documents/{id}/pages/{n}/image`

OCR and model calls run in a background worker, never in the request. The web app lets a reviewer upload many files at once, watch each move through its steps, inspect every field beside the text it came from, fix and approve flagged invoices, search, and compare two documents.

Home | Document | Review | Compare



5. PDF, image, Excel and CSV processing pipelines

All readers return the same internal structure, so the later stages do not care where the text came from.

Input	How it is read
Digital PDF	PyMuPDF, page by page in batches, so a 1,000-page PDF never sits in memory
Scanned PDF, TIFF	OpenCV cleanup (deskew, denoise, threshold), then Tesseract; an upside-down page is detected and turned. Pages with low OCR confidence go to Gemini vision
Phone photo (JPG, PNG)	The same OCR step, then Gemini vision for every photo (D82), with fallbacks to the text model and then the OCR text
Excel (XLSX, XLS)	openpyxl and xlrd, every sheet, header row detection, merged cells
CSV	Python's csv module, with the encoding and delimiter detected
Broken files	Each fails alone with a clear code (`CORRUPTED_FILE` , `ENCRYPTED_PDF` , `EMPTY_FILE` , `UNSUPPORTED_TYPE`) and never stops the batch

Gemini returns each field as printed, with a confidence score and the text it came from, as JSON checked by Pydantic. A missing or unclear value stays empty: the model is told never to guess. Code then parses dates and money, checks that the evidence for the invoice number and total appears in the source, and runs eleven validation rules in `Decimal` arithmetic.

6. Automated tests and test results

TEST_REPORT.md has the full report. From CI run 84 on 5 November 2026:

Job	Result
Backend: ruff, migrations, 428 pytest tests on Postgres 17 with pgvector and Tesseract	Passed, line coverage 96%
Frontend: ESLint, TypeScript, production build	Passed
Secret scan: gitleaks over the full Git history	Passed
Docker: both images build, the compose stack starts, a CSV upload is processed and found by search	Passed

The tests cover document processing, OCR, spreadsheet parsing, AI extraction (with a fake model), validation, every API endpoint, and failure cases: corrupted, encrypted, empty and mislabelled files, model errors, retries, timeouts and worker crashes. CI runs them on every push.

7. Evaluation report with measured metrics

EVALUATION.md has the method, every number and the weak spots. Evaluation run 5 used the real Gemini models on 44 synthetic files with 91 labelled invoices:

Metric	Result
Extraction accuracy	100% of 728 field values (8 fields on 91 invoices)
Line items	224 of 224 (precision and recall 100%)
Document success rate	100% of valid files (35 of 35)
Failure rate	0%
Broken files rejected with the right code	9 of 9
Planted errors caught	12 of 12, with 1 false flag in 80 clean invoices
Files that are not invoices	4 of 4 gave no invoice
Processing time per document	Spreadsheet 1.9 s, digital PDF 23.2 s, scanned PDF 28.3 s (mean)
Cost at paid Gemini prices	About $0.0015 per invoice

Run 8 added nine phone photos at three levels of difficulty: 87.5% of fields right (mild and medium 100%, hard 66.7%). A 1,000-page PDF reads in 2.6 seconds with a 160 MB memory peak. The data is synthetic and cleaner than real invoices, so real accuracy will be lower; the report says where.

8. Recorded walkthrough of the working application

Video: link added before submission.

It shows a mixed upload, live progress, extracted results, a validation flag and its correction, keyword and semantic search, PDF to Excel cross-verification, a failed file and its retry, the API docs, CI and the evaluation report. The script is in WALKTHROUGH.md.

9. AI prompts and disclosure of AI-assisted development

AI_USAGE.md records which AI tools were used, the main prompts of each working session, what AI wrote, and how each part was checked.

- **Tools.** Claude for planning, and Claude Code for writing code, tests and documentation, phase by phase. Gemini is the runtime model inside the product.
- **Division of work.** I made the architecture and scope decisions and reviewed each phase. Claude Code wrote most of the code under those decisions. A phase closed only when its tests passed in CI.
- **Runtime prompts.** Every prompt the product sends to Gemini is in `backend/app/extract/prompts.py`: extraction, one retry with the validation error, spreadsheet column mapping, vision transcription and search embeddings.
- **Safety rules for the model.** It only reads; all arithmetic runs in Python. Document text sits between fixed markers and is treated as data, and the model gets no tools. One test invoice carries a hidden instruction to change its total, and the system still reads the printed value.

Speed, scalability, reliability and security

Quality	What shows it
Smooth performance	Uploads return at once and the work runs in the background, with each file's steps shown live. Home holds 60 frames a second with the CPU slowed four times.
High speed	A digital page reads in about 3 ms and a spreadsheet in about 2 seconds. The windows of a long PDF go to the model in parallel. Most of the remaining time is the free tier's limit of 12 model calls a minute, which is a setting.
Scalability	A 1,000-page PDF streams in batches with flat memory. The job queue lets workers share the load: 1, 2 and 4 workers ran 1.0x, 2.0x and 3.8x as fast.
Reliability	Jobs live in the database and survive a restart. Failed calls retry with backoff, a failing model falls back to a second one, and a broken file fails alone. A health check covers the database and file storage.
Strong security	Secrets live only in the hosting settings, and gitleaks scans the full Git history on every push. Uploads are checked by their bytes. Files sit in a private bucket. Every route except login and health needs a login token, login and upload are rate-limited, and logs never hold document text. See SECURITY.md.

Where the build differs from the suggested stack

The brief allows alternatives with reasons.

Suggested	Used	Why
Node.js or NestJS	Python and FastAPI	The OCR, PDF and spreadsheet libraries are Python. One language means one deployment (D01).
Redis with BullMQ	A job table in PostgreSQL	Jobs commit with the data they change and survive a restart, with no extra service (D02).
A separate vector database	pgvector in PostgreSQL	One query can filter by supplier, date and amount and rank by meaning (D05).
pandas	openpyxl and xlrd	pandas would add about 60 MB and more memory on a 512 MB instance, for no gain (D37).